

NeuronBench recovery by numerical, symbolic, and manual matching

Top-1 run 313196 and top-2 run 313195

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Main result. Across 45 held-out fits, the strict numerical threshold finds 29 recoveries, the existing project symbolic checker finds 33, and manual whole-frontier inspection finds 35. All 29 numerical recoveries are manually valid. The project checker has two manual false positives and four manual false negatives. Manual inspection additionally accepts six scientifically recognizable current-balance laws whose NRMSE is just above 10^{-6} .

Runs and three match criteria

Run **313196** (“top-1”) evolved on `z_rebound` and was evaluated on five other worlds with five seeds each. Run **313195** (“top-2”) evolved on `z_rebound` and `h_sag` and was evaluated on four other worlds with five seeds each. Every criterion searches the complete saved Pareto frontier.

- **NRMSE match:** some row has held-out NRMSE $\leq 10^{-6}$.
- **Project symbolic match:** the current `evaluation.check_pysr_symbolic_match` logic, including the $R^2 > 0.5$ guard, three-decimal float rounding, and constant-difference/ratio rules.
- **Manual match:** after undoing target-RMS scaling and expanding in physical units, some row contains every required current-balance monomial, no material extra monomial, and clearly close fitted coefficients. Tiny floating artifacts are allowed; missing I_{ext} or leak- V is not.

Two inexpensive GPT-5.6 Luna reviewers each inspected one complete evolution run (25 and 20 frontiers). A third Luna reviewer audited all 45 decisions. The primary and audit decisions agreed on 44/45 cases before adjudication. Selected rows were programmatically checked against the correct source run after an initial cross-run cache contamination was detected and corrected.

Aggregate results

Regime	Training worlds	Fits	NRMSE	Project symbolic	Manual
Top-1 (313196)	<code>z_rebound</code>	25	14 (56%)	19 (76%)	19 (76%)
Top-2 (313195)	<code>z_rebound</code> , <code>h_sag</code>	20	15 (75%)	14 (70%)	16 (80%)
Combined		45	29 (64%)	33 (73%)	35 (78%)

The identical top-1 totals for symbolic and manual matching hide a case swap: manual review adds `ca_rebound` seed 10000 but rejects the symbolic-only `d_type` seed 10003.

Regime	Held-out world	Fits	NRMSE	Project symbolic	Manual
Top-1	<code>h_sag</code>	5	0	1	1
	<code>na_fatigue</code>	5	5	5	5
	<code>ca_rebound</code>	5	2	3	4
	<code>d_type</code>	5	3	5	4
	<code>textbook_M</code>	5	4	5	5
Top-2	<code>na_fatigue</code>	5	5	5	5
	<code>ca_rebound</code>	5	4	3	4
	<code>d_type</code>	5	3	4	4
	<code>textbook_M</code>	5	3	2	3

Agreement and disagreement

The table below partitions all 45 fits. A check denotes that the criterion calls the frontier a match.

NRMSE	Symbolic	Manual	Count	Interpretation
✓	✓	✓	26	unanimous recovery
✓	–	✓	3	symbolic-checker false negatives
–	✓	✓	5	recognizable law just above NRMSE threshold
–	–	✓	1	manual-only recognizable law
–	✓	–	2	symbolic-checker false positives
–	–	–	8	unanimous non-recovery

Manual matches above the numerical threshold. Six frontiers have complete current-balance structure and close coefficients but best NRMSE above 10^{-6} :

Run	World	Seed	Best NRMSE	Project symbolic?
313196	<code>h_sag</code>	10004	1.856×10^{-6}	yes
313196	<code>ca_rebound</code>	10000	7.376×10^{-6}	no
313196	<code>ca_rebound</code>	10001	4.282×10^{-6}	yes
313196	<code>d_type</code>	10002	7.455×10^{-6}	yes
313196	<code>textbook_M</code>	10002	2.112×10^{-5}	yes
313195	<code>d_type</code>	10001	5.235×10^{-5}	yes

Symbolic false positives. Both are `d_type`, seed 10003: run 313196 has NRMSE 7.94×10^{-4} and run 313195 has 8.10×10^{-4} . Manual expansion finds a missing or materially inaccurate leak/intercept contribution. Coarse rounding followed by the checker’s constant-difference rule erases the discrepancy.

Symbolic false negatives. Four manually valid frontiers fail the project checker: run 313196 `ca_rebound` seed 10000, plus run 313195 `ca_rebound` seed 10000, `d_type` seed 10004, and `textbook_M` seed 10001. The latter three have NRMSE between 2.5×10^{-10} and 3.1×10^{-7} ; factorization-sensitive rounding, not a scientific mismatch, causes their symbolic failures.

The one reviewer disagreement

The top-1 primary reviewer accepted `ca_rebound`, seed 10000; the audit reviewer rejected it. Canonical expansion gives complete ground-truth support, no extra monomials, and NRMSE 7.376×10^{-6} . Representative physical coefficients are

$$I_{\text{ext}} : 0.999908 (1), \quad V : -0.297741 (-0.3), \quad V\phi_T : -3.200709 (-3.2), \quad \phi_T : 384.027 (384),$$

where parentheses contain ground truth. The largest relative discrepancy among required coefficients is about 0.8% (the leak- V coefficient), so the final manual label is **accept**. This medium-confidence adjudication is retained explicitly in the machine-readable record.

Interpretation

The 10^{-6} NRMSE rule is conservative but clean in this sample: every numerical recovery survives manual inspection. The project symbolic checker increases coverage, but target scaling and representation-dependent rounding create both false positives and false negatives. Manual inspection yields the broadest count (35/45) under the stated physical-law rubric, but it is a judgment-based measure, not a replacement for numerical reporting.

The most transparent summary is therefore to retain all three columns. For an automated future substitute for the manual column, compare canonical expanded polynomials in physical units using scale-aware coefficient tolerances, while requiring every ground-truth variable term to remain present.

Machine-readable judgments and all 45 combined labels: `reports/neuron_manual_match_comparison.json`. Merge and validation script: `scripts/merge_neuron_manual_matches.py`. Source frontiers: `runs/313196/neuron_full_eval/neuron_results.json` and `runs/313195/neuron_full_eval/neuron_results.json`.